HTML & CSS Coding Definitions & Study Guide

Detailed Reference Guide for HTML Elements, CSS Styling, Tables, Forms, Images, and Hyperlinks

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1. Document Structure & Custom Styles (<style>)

HTML provides the baseline structure of a web page, while CSS (Cascading Style Sheets) embedded inside the <style> element controls the visual styling.

- **<!DOCTYPE html>:** Declares the document type as standard HTML5 for modern web browsers.
- **<html> & <head>:** Root element enclosing the document and metadata head section (character encoding, title, internal styles).
- **<style>:** Contains CSS rules. Element selectors (body, h1, table) apply styles globally, class selectors (.song-link) style specific reusable classes, and pseudo-classes (.song-link:hover) handle interactive state changes like mouse hovering.

2. Headings & Paragraphs

Headings define section hierarchy, while paragraphs hold readable body text throughout the document.

- **<h1> to <h6>:** Heading tags establishing visual hierarchy. <h1> represents the primary page title, while <h3> marks sub-topics.
- **<p>:** Defines standard paragraphs of text.
- **Inline CSS (style="..."):** Applies styling rules directly to specific elements (e.g., color, font size).
- **<hr>:** Horizontal Rule tag used to insert a thematic break or divider line between sections.

3. Lists (Unordered & Ordered)

Lists group related items together sequentially or as unordered bullet points.

- ** (Unordered List):** Displays bulleted items without sequential ordering.
- ** (Ordered List):** Displays numbered sequential items.
- ** (List Item):** Defines individual list elements placed inside <ul> or <ol> containers.

4. Tables (<table>)

Tables organize data into structured rows and columns for readable grid displays.

- **<table>:** Parent container defining tabular data structure.
- **<caption>:** Provides an overall header title or summary displayed above the table grid.
- **<tr> (Table Row):** Horizontal container for a single row of cells.
- **<th> (Table Header):** Header cell rendered with bold, centered text by default.
- **<td> (Table Data):** Standard table cell containing individual row data.

5. Hyperlinks (<a>)

Hyperlinks enable navigation between web pages, external websites, or media resources.

- **<a> (Anchor Tag):** Defines clickable text or media links.
- **href Attribute:** Specifies the target URL destination (e.g., YouTube video links).
- **target="_blank":** Opens the destination web page in a new browser window or tab.

6. Images (<img>)

Embeds raster or vector graphics directly into the document flow.

- **<img>:** A self-closing tag used to embed images on a web page.
- **src Attribute:** Specifies the location or online file path of the image.
- **alt Attribute:** Provides alternative descriptive text for accessibility and screen readers.
- **width Attribute:** Sets explicit display dimensions in pixels.

7. Forms & Interactive Inputs (<form>)

Forms collect user-submitted data through interactive inputs, dropdowns, and buttons.

- **<form>:** Container defining interactive controls and submission handling methods (POST, GET).
- **<label>:** Accessible text description associated with specific input controls via the 'for' attribute.
- **<input>:** Primary control with various types: text (single-line input), password (masked input), and submit (action button).
- **<select> & <option>:** Renders a dropdown selection menu with predefined choices.
- **<textarea>:** Provides a multi-line, resizable text box for long descriptions (e.g., hobbies).

8. HTML Code Syntax Corrections Breakdown

Issue Encountered	Original Code	Corrected Code	Explanation
Nested <table> Tag	`<table>` `<table border="1">`	`<table>` `  <tr>...`	Removed duplicate outer table tag preventing double borders.
HTML Tags in Link URL	`href=".../results?search_query=<em>Perfect</em>"`	`href=".../results?search_query=Perfect"`	Removed HTML tags inside the URL string.
Invalid Input Attribute	`<input type="text" id="age" age="age">`	`<input type="text" id="age" name="age">`	Replaced non-standard 'age' attribute with standard 'name' attribute.